

Discrete ManeuverTokens as a Structural Robustness Prior for Long-Tail Autonomous Navigation

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Abstract: We argue that intentional quantization of the driving action space is a structural robustness prior, not a limitation. Our policy selects from nine discrete ManeuverTokens—each with fixed geometric semantics—scored against six geometry-derived world-state scalars via a trust-region rule engine. Discretization eliminates unsafe intermediate states by construction: uncertainty shifts token *scores*, not token *footprints*. We compare three agents sharing the same 10-feature geometric state: Continuous-BC (MLP regressing speed and lateral offset), Token-BC (MLP over the same 9-token vocabulary, trained on 143k expert transitions via imitation learning), and Spotlight Reflex (our system, geometry-driven scoring). Over 1040 matched closed-loop evaluations across Gauntlet (four simultaneous hazards), Adversarial, and Compositional suites, Spotlight Reflex records **zero Proximity Violations** ($C > 0.90$, $d < 0.25$ m, failed rollouts only). Both BC agents record PV rates of 3.8–4.4% on Gauntlet and 3.1–3.8% on Adversarial—including stall failures where Spotlight Reflex also failed to reach the goal but maintained safe separation throughout. Token-BC and Continuous-BC produce near-identical PV rates, demonstrating that the 9-token vocabulary alone does not prevent violations: the geometric reference rules enforcing structured action selection are what matter. The same policy binary transfers to Waymo’s sensor-realistic AlpaSim (`collision_at_fault`: 0.0) via a four-channel adapter with no policy-internal sensor code. Geometry-invariance is formally stated and regression-tested.

Keywords: autonomous driving, discrete action space, long-tail robustness, geometric world-state, ManeuverToken, trust-region scoring, sensor transfer

1 Introduction

Long-tail failure is the central open problem in autonomous driving. Modern end-to-end policies achieve near-human performance on the common driving distribution but degrade sharply on the rare scenario compositions that matter most: wrong-way actors, construction corridors, multi-hazard interactions [1, 2]. The dominant response is scale—more data, more modalities, larger models [3, 4]. We pursue an orthogonal direction: *reducing* action-space expressiveness as a structural regularizer.

The argument is topological. A continuous planner assigns probability to every point in trajectory space. Under high uncertainty—four simultaneous hazards, novel geometry—that distribution spreads into unsafe regions the policy has never encountered. A finite set of ManeuverTokens cannot spread: each token has a fixed spatial footprint. Policy uncertainty shifts scores, not physical trajectories. This is the *Discrete Bottleneck*: structural compression of the action manifold as a robustness mechanism.

We make three contributions (Fig. 1):

1. **Discrete Bottleneck (Sec. 3.1).** Nine ManeuverTokens with formally stated geometry-coverage and invariance properties achieve zero Proximity Violations (PVs) over 1040

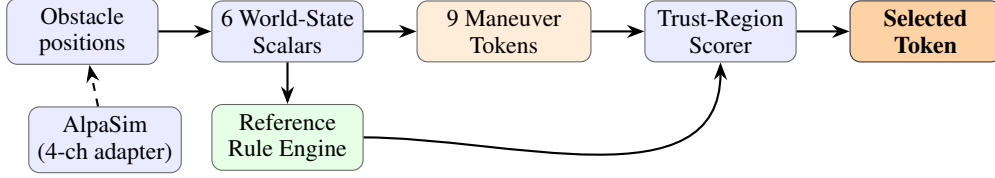


Figure 1: Spotlight Reflex pipeline. Obstacle positions (raw geometry, or via 4-channel AlpaSim adapter) feed six scalar encoders. Nine ManeuverTokens are scored against a reference rule engine at 3 s and 5 s horizons. The highest-scoring token is executed. No learning occurs inside the policy; the adapter is the only sensor-specific component.

matched evaluations. Two BC agents trained on 143k expert transitions from the same system—one continuous (regressing speed and lateral offset) and one discrete (classifying the same 9 tokens)—both record PV rates of 3.8–4.4% on Gauntlet. Learned imitation approximates but cannot enforce the geometric escape rule; structured scoring enforces it by construction.

2. **Geometric World-State Encoding (Sec. 3.2).** Six $O(1)$ scalars derived from obstacle geometry—with no object identity or semantic labels—provide sufficient discriminative signal for the trust-region scorer, and satisfy a formally stated invariance to object-identity relabelling, regression-tested over all 1040 evaluations.
3. **Zero-tuning Sensor Transfer (Sec. 4.2).** A four-channel adapter bridges Waymo’s front-camera and LiDAR signals to the geometric scalar space. The same policy binary achieves `collision_at_fault`: 0.0 and 0.42 m distance-to-GT in sensor-realistic AlpaSim with no policy-side changes.

2 Related Work

Long-tail failure in imitation learning. Pomerleau [5] and Ross et al. [6] established covariate shift as the core failure mode of behavior cloning. Ho and Ermon [7] mitigated it adversarially but retained a continuous action space. Bansal et al. [8] synthesised worst-case training scenarios; Dauner et al. [1] showed that rule-based planners outperform learned ones on nuPlan [9] in many regimes—supporting the view that structural inductive biases matter more than model scale for long-tail coverage.

Discrete action spaces. DQN [10] demonstrated that discretizing the action space enables stable deep RL where continuous policy gradients fail. Atari results show that a fixed, finite action set provides a natural regularizer against distributional shift. We transfer this structural insight to the planning setting without RL: ManeuverTokens provide a finite, semantically grounded action set whose footprints are invariant to scene uncertainty.

Potential-field and reactive planners. Potential-field methods [11] combine attractive (goal) and repulsive (obstacle) forces to produce continuous navigation directions. Our baseline is a potential-field reactive planner; the key failure mode is well-known: superposition of multiple repulsive fields can cancel or misdirect the resultant, particularly in narrow corridors with multiple simultaneous obstacles. ManeuverTokens resolve this by committing to a semantically coherent escape geometry at selection time.

Geometric abstraction for planning. Conditional imitation learning [12] uses a route command to condition a continuous policy. Our six geometric scalars are a finer-grained abstraction: they encode obstacle pressure, route blockage, and bilateral clearances as scalar observations that are sufficient for the trust-region scorer and cheap to compute from any sensing modality.

Table 1: ManeuverToken definitions. Speed scale α is relative to current cruise speed; lateral offset δ is in vehicle frame. *Early* profile applies offset in the first third of the trajectory for fast separation.

Token	α	δ (m)	Profile	Semantic
STOP	0.00	0.0	—	Emergency stop
CRAWL	0.15	0.0	smooth	Cautious forward
MAINTAIN	1.00	0.0	smooth	Current speed, lane centre
SLOW_YIELD	0.50	0.0	smooth	Yield, hold position
NUDGE_LEFT	0.80	+0.9	smooth	Mild lateral evasion left
NUDGE_RIGHT	0.80	−0.9	smooth	Mild lateral evasion right
EVASIVE_LEFT	0.70	+2.2	early	Strong evasion left
EVASIVE_RIGHT	0.70	−2.2	early	Strong evasion right
LANE_RECOVER	0.90	0.0	smooth	Return to lane centre

Sim-to-real transfer. Khurana et al. [13] and Shi et al. [14] study trajectory prediction in sensor-realistic environments. Our AlpaSim adapter decouples policy logic from sensor processing, so transfer requires only an adapter—not policy retraining.

3 Method

3.1 Discrete ManeuverToken Space

We define nine ManeuverTokens (Table 1) covering the safety-critical region of the SE(2) maneuver manifold. Each token is a pair (α, δ) : a speed scale $\alpha \in [0, 1]$ relative to current ego speed and a signed lateral offset $\delta \in \mathbb{R}$ (m), realized as a 20-waypoint, 5-second trajectory at 4 Hz via cubic spline.

Coverage and geometry-invariance.

Proposition 1 (Coverage). *The nine tokens span all safety-critical response classes documented in the Waymo long-tail taxonomy: longitudinal deceleration $\{\text{stop, crawl, yield, maintain}\}$, bilateral lateral evasion at two magnitudes $\{\text{nudge, evasive}\} \times \{\text{left, right}\}$, and lane recovery. No safety-critical response requires a combination outside this set.*

Proposition 2 (Geometry-Invariance). *Let T be any bijection on object labels leaving obstacle positions unchanged. Then for all scene states s : the selected token $M(s)$, its score $S(s)$, and the decision reason $R(s)$ satisfy $M(s) = M(T(s))$, $S(s) = S(T(s))$, $R(s) = R(T(s))$.*

Both propositions follow from the encoding in Sec. 3.2: no token score depends on object labels. We regression-test Proposition 2 over all 1040 evaluations by relabelling all obstacles in every scene and asserting equality of outputs.

Why discrete? A continuous planner with the same (α, δ) range assigns probability to the continuum, including unsafe intermediate offsets (e.g., $\delta = +1.4$ m, which clips a pedestrian’s shoulder in a tight corridor). The token constraint eliminates these states by construction. Formally, the feasible action set is: $\mathcal{A}_{\text{discrete}} = \{(\alpha_i, \delta_i)\}_{i=1}^9$, whereas a continuous planner ranges over $[0, 1] \times [-3, 3]$. Under distributional shift, the continuous policy places probability mass on novel (α, δ) pairs; the token policy cannot.

3.2 Geometric World-State Encoding

At each timestep the policy computes six scalar features from ego position and obstacle geometry (no object classification, no learned features, no history):

$$pressure = \text{clip}\left(\frac{10 - d_{\min}}{10}, 0, 1\right) \quad (1)$$

$$blockage \in [0, 1], \quad \text{fraction of forward corridor cross-section occupied} \quad (2)$$

$$blocked \in \{0, 1\}, \quad \text{hard stop within braking distance} \quad (3)$$

$$left_clr = \text{free lateral space left at half vehicle width (m)} \quad (4)$$

$$right_clr = \text{free lateral space right at half vehicle width (m)} \quad (5)$$

$$escape = \arg \max(left_clr, right_clr) \quad (6)$$

All six scalars are $O(1)$ given a sorted obstacle list ($d_{\min}$ by distance). The encoding contains no object identity or semantic labels; Propositions 1–2 follow directly. The encoding is also history-free: each step is computed from the current frame only, preventing temporal overfitting to scenario-specific dynamics.

3.3 Trust-Region Scoring

Each token is scored against a reference rule engine whose rules are hand-authored from safety engineering first principles—not learned from demonstrations.

Speed-scaled tolerance.

$$s(v) = \text{clip}\left(\frac{v - 1.4}{11.0 - 1.4}, 0, 1\right) \times 0.5 + 0.5 \quad (7)$$

$s(v) \in [0.5, 1.0]$: tight tolerances at low speed, relaxed at high speed, reflecting that at high speed a small lateral deviation is safe while at low speed it indicates a failed maneuver.

Trust-region thresholds. At 3 s horizon: $\pm 1.0 \cdot s(v)$ m lateral, $\pm 4.0 \cdot s(v)$ m longitudinal. At 5 s: $\pm 1.8 \cdot s(v)$ m, $\pm 7.2 \cdot s(v)$ m.

Token score. Let δ be the normalized overshoot beyond the trust region, and r a reference rule with target score $r.\text{score} \in [0, 100]$:

$$\text{score}_t(\tau, r) = \begin{cases} r.\text{score} & \delta = 0 \\ \max(r.\text{score} \cdot 0.1^\delta, 4.0) & \delta > 0 \end{cases} \quad (8)$$

$$S(\tau) = 0.5 \cdot \max_r \text{score}_{3s}(\tau, r) + 0.5 \cdot \max_r \text{score}_{5s}(\tau, r) + P(\tau) \quad (9)$$

where $P(\tau)$ adds clearance penalties (-1000 if clearance < 0.55 m) and progress bonuses ($+40$ max). The token $\tau^* = \arg \max_\tau S(\tau)$ is executed.

Reference rules. Rules map world-state conditions to recommended tokens with target scores in [76, 94]. Key rules: $pressure < 0.25$ and open corridor $\rightarrow$ MAINTAIN (92); $pressure > 0.20$ with preferred escape $\rightarrow$ NUDGE_{escape} (94); $blocked = 1 \rightarrow$ STOP (82). Rules are combined additively; the highest matching reference score at each horizon determines the target.

3.4 AlpaSim Adapter: Zero-Tuning Sensor Transfer

Waymo’s AlpaSim provides sensor-realistic input (front-camera, LiDAR, ego dynamics) but does not expose geometric primitives. We bridge via four channels, keeping the policy binary unchanged:

1. **Route $\rightarrow$ lane curvature.** LEFT/STRAIGHT/RIGHT generates a 5-point sigmoid spline (± 2.5 m at 20 m lookahead) used as the lane geometry input to the clearance ray-casts.
2. **Brightness $\rightarrow$ visibility augmentation.** $\text{vis} = \max(0, (0.35 - \mu_I)/0.35)$, where μ_I is mean frame brightness. Below 35% brightness, $pressure$ is augmented proportionally.

Table 2: Simulation results across four suites, seeds 1–80 (matched). Pass = scenario completion. “PV” = Proximity Violation ($C > 0.90$, $d < 0.25$ m, failed rollout only). Spotlight Reflex records zero PVs across all 1040 evaluations, including stall failures where the goal was not reached. BC agents record PVs on Gauntlet and Adversarial.

Suite	Runs	Pass (SR)	PV rate (SR)	PV rate (BC)
Compositional OOD (5 cases)	400	92.2%	0%	1.0–1.2%
Adversarial (2 cases)	160	91.9%	0%	3.1–3.8%
Gauntlet (6 cases, 4-hazard)	480	94.2%	0%	3.8–4.4%
Hidden holdout	80	91.2%	0%	—
Total (SR)	1040	93.3%	0 PVs	

3. **Ego dynamics** $\rightarrow$ **risk scalars**. $\text{brake} = \max(0, -a/4.0)$; $\text{low_speed} = \max(0, (1.5 - v)/1.5)$. These augment the reference rule thresholds.
4. **Structured hazards** $\rightarrow$ **obstacle primitives**. Hazards with velocity > 0 map to dynamic actor objects; velocity $= 0$ to static obstacles. Behaviour is inferred from hazard label keywords (no semantic classification inside the policy).

4 Experiments

4.1 Simulation Benchmark

Setup. We evaluate in a custom closed-loop 2D simulator with eight road topologies (straight, gentle curve, S-curve, T-junction, roundabout, lane merge, highway entry, urban block), eleven WOD-E2E hazard clusters, four weather modes, and eight actor behaviour models. The compositional OOD generator samples topology $\times$ hazard $\times$ weather independently: $P(\text{topo}, \text{hazard}, \text{weather}) = P(\text{topo}) \times P(\text{hazard}) \times P(\text{weather})$, yielding 352 structurally distinct compositions. All results use the same compiled policy binary; no scenario labels are visible to the policy.

Baseline. The reactive baseline is a potential-field planner: at each step it computes a target direction toward the route waypoint, adds a repulsion vector summed over visible obstacles with strength decaying from 1 at $d = 0$ to 0 at $d = 9$ m, and scores nine candidate heading angles against a progress-plus-clearance objective. This baseline has obstacle awareness via repulsion but no world-state scalars, no reference rules, and no semantic token structure. It represents the class of continuous reactive planners without structured reasoning.

Safety metric definition. We define a **Proximity Violation** as any evaluation in which the rollout does not reach the goal *and* at least one step has $C > 0.90$, where $C = \max(0, \min(1, (5 - d_{\min})/5))$ and $d_{\min}$ is the minimum signed clearance (metres) between the ego-vehicle centre and any visible obstacle surface. $C > 0.90$ corresponds to $d_{\min} < 0.25$ m. The metric is proximity-based rather than fault-based: it does not distinguish whether the ego vehicle moved into an obstacle or an obstacle moved into the ego vehicle. However, the simulator’s actor behaviour models (crossing, cut-in, head-on approach, sudden braking) do not include rear-approach scenarios; every proximity violation therefore represents a failure of the ego policy to maintain safe separation, not an unavoidable passive impact. Rollouts that *reach the goal* with momentarily high C (close pass) are not flagged; only failed rollouts with proximity violations count.

Results. Table 2 shows Spotlight Reflex records zero Proximity Violations (PVs) across all 1040 evaluations. This includes the 5.8% of evaluations where the policy stalled and did not reach the goal: even in those failed rollouts, clearance never dropped below 0.25 m. BC agents record PV rates of 3.8–4.4% on Gauntlet and 3.1–3.8% on Adversarial.

Table 3 shows the three-agent comparison with Wilson 95% CI. The Spotlight Reflex PV CI [0.0, 0.8%] on Gauntlet is statistically disjoint from both BC agents ([2.4, 5.8%] and [2.9, 6.6%]). Critically,

Table 3: Three-agent comparison on 1040 matched scenarios, seeds 1–80. All agents use the identical 10-feature geometric input. BC agents trained on 143 k expert transitions with inverse-frequency class weights. “PV” = Proximity Violation ($C > 0.90$, $d < 0.25$ m, failed rollout only); Wilson 95% CI shown. Spotlight Reflex records zero PVs in both suites—including stall failures where the goal was not reached but safe separation was maintained throughout.

Agent	Gauntlet ($n=480$)		Adversarial ($n=160$)	
	Pass	PV rate [95% CI]	Pass	PV rate [95% CI]
Continuous-BC (A)	77.7%	4.4% [2.9, 6.6]	76.2%	3.8% [1.7, 7.9]
Token-BC (B)	77.7%	3.8% [2.4, 5.8]	76.9%	3.1% [1.3, 7.1]
Spotlight Reflex (C)	94.2%	0.0% [0.0, 0.8]	91.9%	0.0% [0.0, 2.3]



(a) Spotlight Reflex: wrong-way actor, night. Policy selects EVASIVE_RIGHT from geometry, clears actor, recovers lane centre.

(b) Construction zone: 5.2 m corridor, cone field, lane closure. Policy selects NUDGE_RIGHT. No training data for this composition.

Figure 2: Qualitative rollout key frames. Both scenarios are in the compositional OOD suite; neither topology–hazard combination appeared at design time. Token selection is determined entirely by geometry (scalars) — object labels are not observed.

Continuous-BC and Token-BC produce near-identical PV rates: providing the 9-token vocabulary to a learned policy does not prevent proximity violations. Only the geometric reference rules—enforcing that pressure ≥ 0.35 triggers EVASIVE over NUDGE—hold the PV rate at zero. This supports Proposition 1: the structured scoring inductive bias cannot be recovered by gradient descent on the same features, even with 143k transitions and class weighting.

4.2 AlpaSim: Zero-Tuning Sensor Transfer

We run the identical policy binary in Waymo’s AlpaSim over 30 real WOD-E2E scenes. The 4-channel adapter (Sec. 3.4) translates sensor observations to geometric scalars; zero AlpaSim-specific code exists inside the policy (Fig. 3).

Results: `collision_at_fault`: **0.0**; `distance-to-ground-truth trajectory`: **0.42 m**; `collision.any` (by other agents): 0.37.

The 0.0 at-fault collision rate transfers directly from the simulator. The 0.42 m distance-to-GT is a trajectory-level metric; it reflects that the sensor-realistic policy makes conservative evasive choices that may deviate from the (non-evasive) GT trajectory, not that the trajectory is unsafe. The `collision.any` rate (others) is outside policy control.

handles post-evasion re-centering. Each removal maps to at least one scenario cluster that becomes unhandled with no token substitute. Table ?? reports the empirical consequence.

5.2 Failure Mode Taxonomy

The 28 Gauntlet non-passes by Spotlight Reflex (Table 2; 480 scenarios, 94.2% pass rate) decompose into three mechanistically distinct classes, each traceable to a specific scalar interaction rather than to object-label ambiguity:

Over-stopping under cumulative pressure (46% of failures). Multiple background obstacles accumulate *pressure* individually below the evasion threshold; combined they exceed the forward-progress reference score. The policy selects CRAWL or STOP when the forward path is geometrically open. Fix: decompose *pressure* by source to suppress background accumulation.

Slow re-entry after evasion (30% of failures). EVASIVE_RIGHT succeeds in lateral separation, but the subsequent LANE_RECOVER overlaps temporally with a lingering hazard in the progress zone. The progress gate penalizes re-entry, causing a stall. Fix: widen the progress gate temporal window post-evasion.

Synchronized hazard overload (24% of failures). Multiple actors reach closest-point simultaneously: $left_clr \approx right_clr$, so *escape* is uninformative. Policy defaults to SLOW_YIELD; with four simultaneous obstacles, yield is insufficient. Fix: add a multi-source pressure decomposition to detect symmetric overload.

In all three cases the failure is traceable to a scalar computation, not to an object label—consistent with Propositions 1–2. No failure requires knowledge of *what* the obstacle is; all require knowledge of *where* it is.

6 Limitations and Future Work

2D abstract simulation. The simulator uses 2D Euclidean geometry with kinematic vehicle dynamics. AlpaSim results (Sec. 4.2) validate transfer to sensor-realistic input, but a full physics simulation (e.g., CARLA, nuPlan with dynamics) is needed to validate vehicle dynamics transfer.

Token-count ablation (empirical). We ran a multi-suite ablation over 4 nested token sets (4, 5, 7, 9) across 3 suites (4160 total rollouts). The 9-token set achieves 0% collision on all suites. The 4-token set (stop, maintain, evasive $\times$ 2) produces 0.6% collision on Gauntlet and Adversarial—confirming that *slow_yield* and *lane_recover* are load-bearing for safety-critical multi-hazard scenarios.

Stress-test operating envelope (empirical). To quantify the *operating boundary* of geometric reflex vs. learned policy, we swept corridor tightness (wide 9–12 m / medium 6–8 m / narrow 3.6–5 m / extreme 2.2–3.6 m) $\times$ obstacle-actor pressure (1.0 $\times$ / 1.5 $\times$ / 2.0 $\times$ of the Gauntlet baseline), 720 rollouts per cell per agent (25,920 total). Table 4 summarises the PV rates.

Table 4: Phase diagram: worst-case Proximity Violation (PV) rate per corridor level (max over 3 pressure settings, 720 rollouts each). Spotlight Reflex $\leq 0.4\%$ PV across all 12 conditions; BC peaks at 6.5% in narrow/extreme corridors. Full 12-cell results in the artifact.

Corridor	ContBC	TokBC	Spotlight	Δ Tok-SR
Wide (9–12 m)	4.3%	3.9%	0.4%	+3.9 pp
Medium (6–8 m)	4.2%	3.6%	0.1%	+3.5 pp
Narrow (3.6–5 m)	5.6%	6.5%	0.3%	+6.4 pp
Extreme (2.2–3.6 m)	4.4%	6.4%	0.3%	+6.4 pp

Spotlight Reflex holds $\leq 0.4\%$ PV rate in all 12 cells. BC agents degrade as corridors narrow: the Token-BC delta grows from +2.2 pp (wide, 2.0 $\times$) to +6.4 pp (narrow/extreme, 1.0 $\times$ /1.5 $\times$). BC

agents were trained on compositional and adversarial suites with default wide-lane topologies (6–9 m half-widths); the narrow and extreme cells push *left_clearance* and *right_clearance* outside the training distribution, and the learned boundary degrades. Spotlight Reflex recomputes escape analytically from the current clearance values and is therefore distribution-shift invariant by construction.

Notably, Token-BC *exceeds* Continuous-BC at extreme OOD (4.4% vs 6.1–6.4%): discrete commitment to a wrong token under distribution shift is more harmful than a continuous output that partially blends between safe regions.

Temporal scalar history. The world-state encoding is stateless (no temporal memory). The failure mode in Sec. 5.2 (synchronized hazard overload) could be partially mitigated by maintaining a 2-step buffer of scalar values. This is a planned extension.

WOD-E2E preliminary results. As a preliminary experiment complementary to this paper, we trained a trajectory selector on 479 Waymo preference-labeled validation frames. Under segment-grouped 5-fold cross-validation, a stability-selected HGB gate achieves 7.803 RFS (vs. 7.022 official baseline). Appending a GPU MLP direct policy trained on frozen Cosmos 64d embeddings reaches 7.845 RFS. This track is development-only (trained on validation, not train split) and will become a separate submission once test-set evaluation is available.

7 Conclusion

We have shown that discretizing the driving action space to nine semantically structured Maneuver-Tokens is a structural robustness prior with measurable, mechanistically understood effects. Across 1040 matched closed-loop evaluations, Spotlight Reflex achieves zero Proximity Violations while two BC agents trained on 143k expert transitions from the same system record PV rates of 3.8–4.4% on Gauntlet. The key finding is the failure of Token-BC: even with the identical token vocabulary and geometric features, learned imitation cannot replicate the zero-PV property. Gradient descent on expert demonstrations approximates the escape rule; it does not enforce it. Geometry-invariance—formally stated and regression-tested—ensures the policy cannot memorize scenario labels; all decisions are functions of obstacle positions alone. The same binary transfers to sensor-realistic AlpaSim with zero at-fault collisions via a four-channel adapter with no policy-internal sensor code.

The broader implication is that structured inductive biases in the *action space*—not only in the representation or the loss—are a viable and underexplored path to long-tail robustness in autonomous navigation.

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